



Image-Based Ship Classification

Using Deep Learning Techniques for Intelligent Maritime Monitoring

A Comparative Study of CNN and Vision Transformer Architectures
on the FGSC-23 Satellite Imagery Dataset

Group 4:

Amreen Kaur • Narshini Chuckowry • Kshitij Santokhee • Bhavya Bhola

Thompson Rivers University | Kamloops, BC

02 Meet the Team



Amreen Kaur

Computer Science



Narshini Chuckowry

Data Science



Kshitij Santokhee

Computer Science



Bhavya Bhola

Computer Science

03 Introduction & Background

Why automated ship classification?

Over 80% of global trade moves by sea, yet maritime traffic is still largely monitored by hand.

Traditional systems rely on radar and AIS signals - these fail when signals are weak, spoofed, or absent.

High-resolution satellite and aerial imagery offers a signal-independent alternative, but fine-grained ship classification is hard: small targets, varying scale, sea clutter, and visually similar vessel types (e.g., destroyers vs. frigates).

Our contribution

A head-to-head comparison of 6 pretrained CNNs and 3 Vision Transformers on the FGSC-23 fine-grained ship dataset, under an identical training pipeline.

23

Fine-grained ship classes

Military + civilian vessels

4,080

High-resolution satellite images

Google Earth + Gaofen-2

9

Deep learning models compared

6 CNNs + 3 ViTs

86%

Best validation accuracy

ViT-B/16 - our top model

04 Literature Review

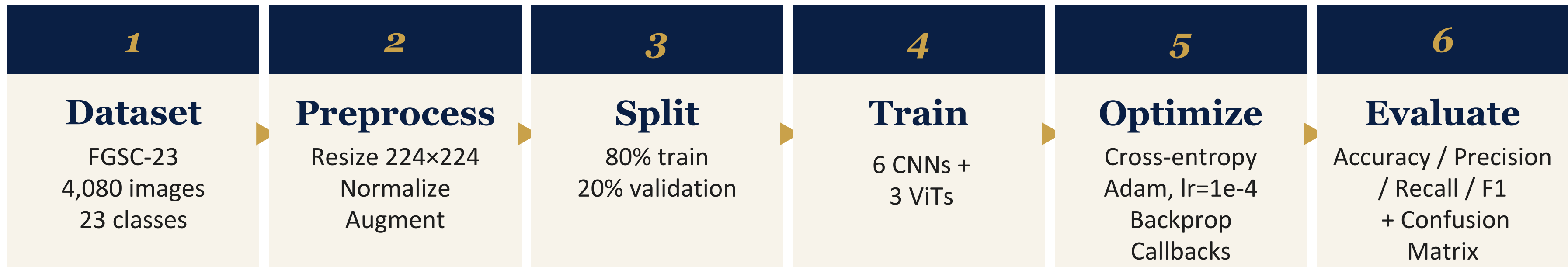
A survey of recent ship detection and classification methods

Reference	Method	Dataset (Images)	Accuracy
Huang et al. [7]	XGBoost + Random Forest (AIS data)	AIS ship trajectories (9,785)	97%
Escorcia-Gutierrez et al. [8]	OMRCNN-SHD (Mask R-CNN + WRELM)	250 ship + 1,000 non-ship (augmented)	98.63%
Mazen & Mazen [9]	Cascaded Mask R-CNN	Airbus Ship Detection (192,000)	98.4%
Zheng et al. [10]	Msf-AT (multi-scale attention CNN)	OpenSARShip	+3.97%
Singh et al. [11]	Hybrid Active Contour + YOLOv8	SSDD (1,160 SAR images)	98.7%
Dong et al. [15]	Multiscale Contrastive Learning Network	FGSC-23 (4,080) / FGSC-41 (9,057)	91.5% / 85.18%
Huang et al. [26]	Ln-SCNet (lightweight CNN)	FUSAR-Ship	76.1%

Research gap: most prior work focuses on binary ship detection or SAR imagery - few studies compare modern CNNs vs. Vision Transformers for fine-grained 23-class optical classification.

05 Methodology & Workflow

Shared pipeline — fair comparison across all 9 architectures



CNN BRANCH

Hierarchical feature extraction via stacked convolutions

VGG16 • VGG19 • ResNet50
MobileNetV2 • InceptionV3 • DenseNet201

Trained 20 epochs | Transfer learning (ImageNet weights)

VISION TRANSFORMER BRANCH

Global self-attention across 16×16 patch embeddings

ViT-B/16 • DeiT-Base • Swin Transformer Tiny

Trained 10 epochs | Pretrained on ImageNet-21k

06 Best Performing Model - ViT-B/16

Vision Transformer Base, 16×16 patches • pretrained on ImageNet-21k

ARCHITECTURE FLOW

Input Image

224 × 224 × 3 RGB satellite image

Patch Partition

Split into 14×14 = 196 patches of 16×16

Linear Projection + [CLS]

Flatten → project into 768-dim embeddings

Transformer Encoder × 12

Multi-Head Self-Attention (12 heads) + MLP

Classification Head

[CLS] token → Linear → Softmax → 23 classes

KEY SPECIFICATIONS

Patch size	16 × 16 pixels
Sequence length	196 patches + 1 [CLS]
Embedding dim	768
Encoder layers	12
Attention heads	12
MLP dim	3,072
Optimizer	Adam, lr = 1e-4
Loss	Categorical Cross-Entropy

Why it wins on FGSC-23

Global self-attention lets every patch look at every other patch - capturing long-range structure across the whole vessel.

Ships vary in scale, orientation, and sea context - attention beats local receptive fields here.

Pretraining on ImageNet-21k gives strong priors that transfer well despite limited training data.

07

Dataset - FGSC-23

Fine-Grained Ship Classification from high-resolution satellite imagery

4,080

HIGH-RES IMAGES

23

SHIP CATEGORIES

224 * 224

INPUT RESOLUTION

80/20

TRAIN / VAL SPLIT

23 vessel categories

MILITARY (IDs 2–9)	SPECIAL & AUXILIARY (IDs 0, 10–13)	CIVILIAN / CARGO (IDs 14–22)
Destroyer, Landing craft, Frigate, Amphibious transport dock, Cruiser, Tarawa-class amphibious assault ship, Amphibious assault ship, Command ship	Non-ship, Submarine, Medical ship, Combat boat, Auxiliary ship	Container ship, Car carrier, Hovercraft, Bulk carrier, Oil tanker, Fishing boat, Passenger ship, Liquefied gas ship, Barge

Dataset challenges

- Class imbalance - some IDs have very few samples
- Varying lighting, angles and sea conditions
- Background clutter near ports and coastlines
- Visually similar classes (destroyer vs. cruiser)
- Some labels noisy or mislabelled

Imagery source

High-resolution satellite imagery captured by Google Earth and Gaofen-2 (GF-2) - manually annotated by domain experts. Sourced from Kaggle.

08 Results - Model Comparison

Evaluation on the FGSC-23 validation set — ViTs clearly outperform CNNs

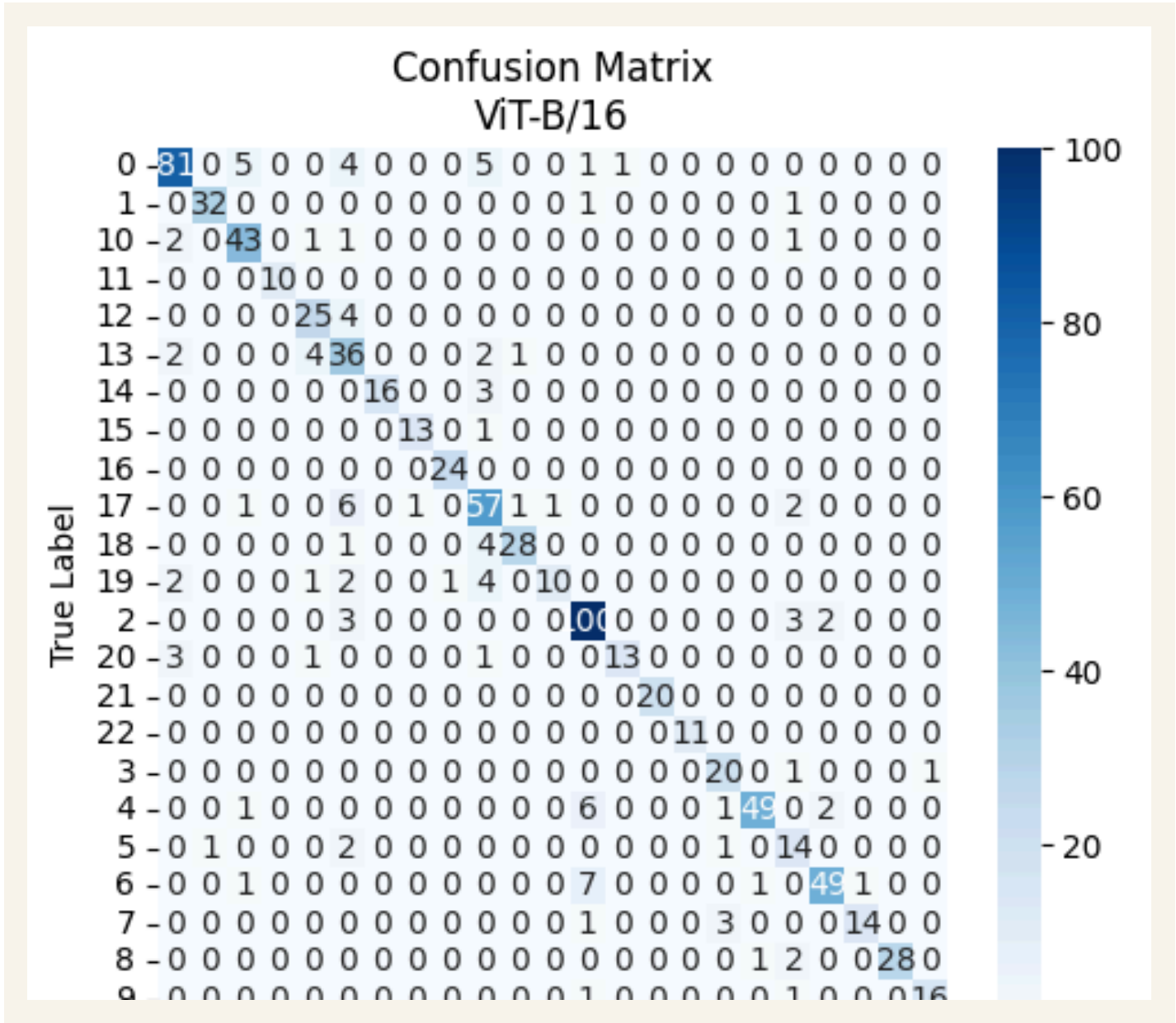
Model	Accuracy	Precision	Recall	F1
VGG16	0.697	0.691	0.688	0.677
ResNet50	0.359	0.337	0.289	0.280
MobileNetV2	0.619	0.698	0.601	0.603
VGG19	0.468	0.484	0.385	0.359
InceptionV3	0.660	0.683	0.644	0.653
DenseNet201	0.712	0.726	0.674	0.686
DeiT-Base	0.689	0.726	0.689	0.662
Swin-Tiny	0.851	0.868	0.851	0.855
ViT-B/16 ★	0.860	0.873	0.860	0.862

★ Best performer across all metrics | Bold = best within CNN / ViT branch

Key takeaway

ViT-B/16 beats the best CNN (DenseNet201) by 14.8 points of accuracy - a 20% relative gain - confirming that global self-attention transfers better to fine-grained satellite ship imagery than purely local convolutions.

Confusion Matrix - ViT-B/16



Strong diagonal → correct predictions across most of 23 classes

09 Work Distribution & Learnings

Team contributions

Member	Responsibility
Amreen Kaur	CNN model implementation (VGG16/19, ResNet50, DenseNet201) and training loops
Narshini Chuckowry	Vision Transformer models (ViT-B/16, DeiT, Swin-Tiny), fine-tuning and hyperparameter search
Kshitij Santokhee	Dataset acquisition, preprocessing pipeline, augmentation strategy and train/val splits
Bhavya Bhola	Evaluation metrics, confusion matrix visualizations, and literature review synthesis
All members	Report writing, results analysis, presentation, and proofreading

What we learned

Architecture matters

Modern ViTs can dramatically outperform classical CNNs when the task needs global context.

Fair comparison is hard

Matching preprocessing, optimizer and epochs is essential for any claim about model superiority.

Data quality > model size

Class imbalance in FGSC-23 hurt even the best models - more balanced data would lift everything.

Transfer learning wins

ImageNet-21k pretraining gave ViT-B/16 a strong head start despite our modest dataset size.

Evaluation beyond accuracy

Confusion matrices revealed which classes were confused - accuracy alone hides this.

10 Conclusion & Future Work

Summary

We benchmarked 6 CNNs and 3 Vision Transformers on the FGSC-23 fine-grained ship dataset under an identical pipeline. ViT-B/16 achieved 86.0% accuracy and an F1 of 0.862, outperforming the best CNN (DenseNet201, 71.2%) by ~15 % - evidence that global self-attention generalizes better than local convolutions for fine-grained optical maritime imagery.

Key contributions

- First head-to-head comparison of six pretrained CNNs and three ViTs on FGSC-23 under a shared pipeline.
- Demonstrated that Vision Transformers significantly outperform CNNs for fine-grained satellite ship imagery.
- Identified specific confusable class pairs via confusion-matrix analysis - guiding targeted future work.
- Reproducible training recipe for the maritime-surveillance community.

Future directions

- Explore larger transformer variants (ViT-L, Swin-B) and hybrid CNN-ViT models for further gains.
- Address class imbalance via focal loss, oversampling and synthetic data from generative models.
- Fuse visual features with AIS trajectory data for robust multi-modal ship classification.
- Distill ViT-B/16 into a lightweight model for real-time deployment on drones and edge devices.

11 References

Key references cited in this study (full list given in the project report)

- [1] Qiao, D., Liu, G., Lv, T., Li, W., & Zhang, J. (2021). Marine vision-based situational awareness using discriminative deep learning: A survey. *J. Marine Sci. & Eng.*, 9(4), 397.
- [7] Huang, I. L., Lee, M. C., Nieh, C. Y., & Huang, J. C. (2023). Ship classification based on AIS data and machine learning methods. *Electronics*, 13(1), 98.
- [8] Escorcia-Gutierrez, J., et al. (2022). Intelligent deep learning-enabled autonomous small ship detection and classification model. *Computers & Electrical Engineering*, 100, 107871.
- [9] Mazen, F., & Mazen, A. (2025). Airbus Ship Classification, Detection and Segmentation using Cutting-Edge Deep Learning Techniques. *Fayoum Univ. J. Engineering*, 8(1), 68–78.
- [11] Singh, T., Babu, T., Nair, R. R., & Duraisamy, P. (2024). Ship detection in synthetic aperture radar imagery: An active contour model approach. *Procedia Computer Science*, 235, 1793–1802.
- [15] Dong, S., Feng, J., & Fang, D. (2024). A novel multiscale contrastive learning network for fine-grained ocean ship classification. *IEEE JSTARS*, 17, 9989–10005.
- [26] Zhao, S., Li, W., Shen, F., & You, M. (2025). Ln-SCNet: A lightweight convolutional neural network for SAR ship classification. *IEEE Access*.
- [27] Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. *CVPR*, 4700–4708.
- [30] Dosovitskiy, A., et al. (2020). An image is worth 16×16 words: Transformers for image recognition at scale. *arXiv:2010.11929*.
- [31] Vaswani, A., et al. (2017). Attention is all you need. *NeurIPS*, 30.

End of presentation slides

